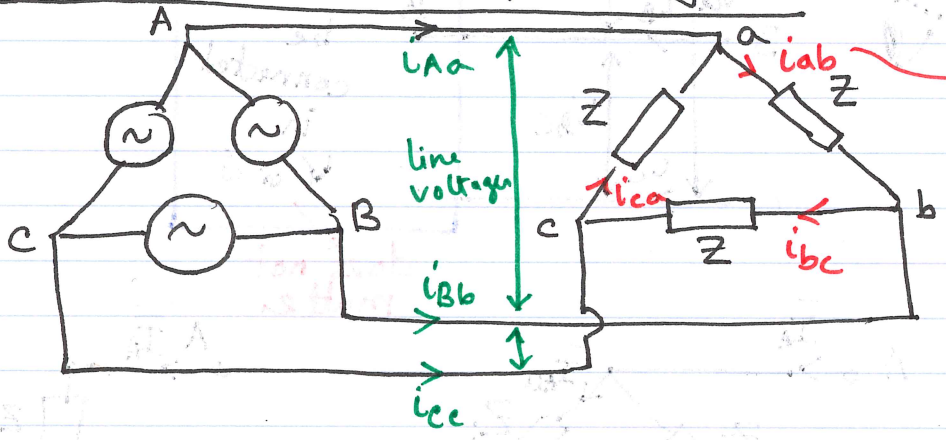
Lecture 7.

Suggested Reading: 3.1/3.2

Last time: * Delta connected three-phase systems.
- Phasors.

Today: * Power equivalence in delta-connected system.
- Approach to attack Δ-systems.
- Typical power system. - where do we use Y, Δ connections.
- Magnetics.

A. Delta-connected three-phase system:



Let us compute per-phase complex power:

$$\bar{S}_{perphase} = \bar{V}_{ab}\,\bar{I}_{ab}^{*}$$ ← note both phase quantities.

or, $P_{phase} + jQ_{phase} = |\bar{V}_{ab}||\bar{I}_{ab}|\cos\phi + j|\bar{V}_{ab}||I_{ab}|\sin\phi$.

∴ Total three phase power,

$$P_{tot} = 3|\bar{V}_{ab}||\bar{I}_{ab}|\cos\phi$$ ← expressed using phase qty.

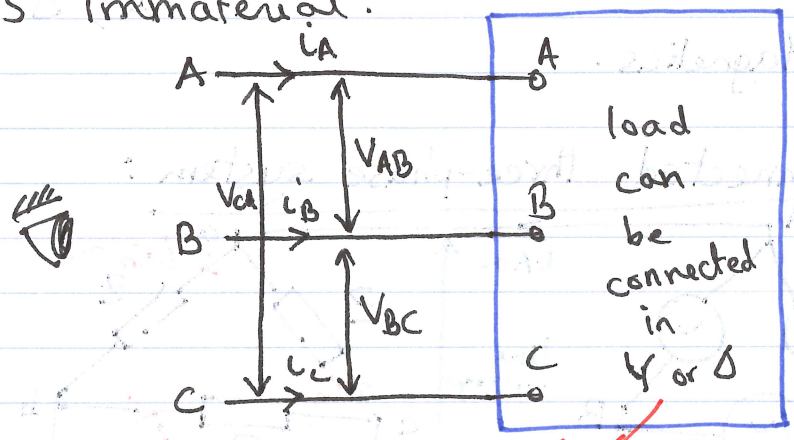
$$= \sqrt{3}\,|\bar{V}_{ab}||\bar{I}_{Aa}|\cos\phi$$ ← expressed using line quantities

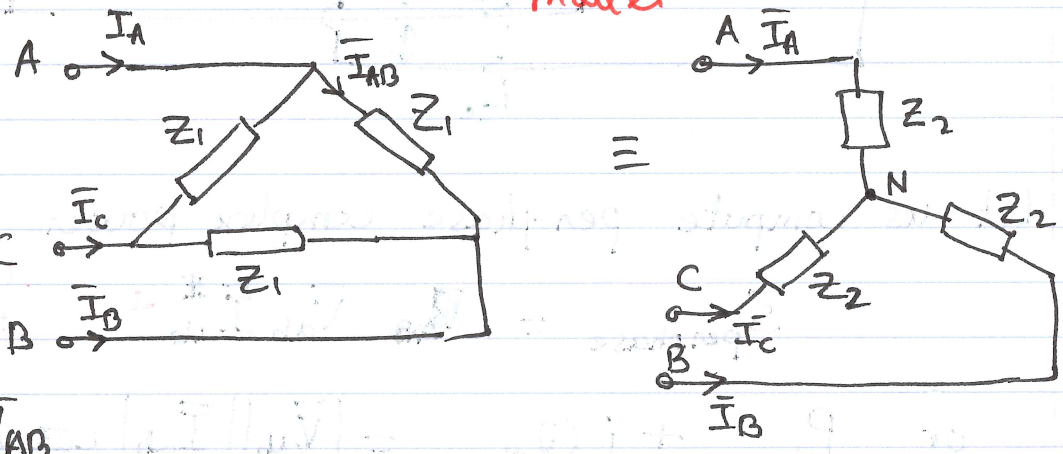
$|\bar{I}_{ab}| = \dfrac{|\bar{I}_{Aa}|}{\sqrt{3}}$

Similarly, total reactive power,

$$Q_{tot} = \sqrt{3}\,|\bar{V}_{ab}||\bar{I}_{Aa}|\sin\phi$$

Therefore, from a ~~reactive~~ power perspective, in terms of line quantities, Y or Δ connection is immaterial.





$$\bar{I}_{AB} = \frac{\bar{V}_{AB}}{Z_1}$$

$$\therefore \quad \bar{I}_A = \sqrt{3}\,\frac{\bar{V}_{AB}}{Z_1}$$

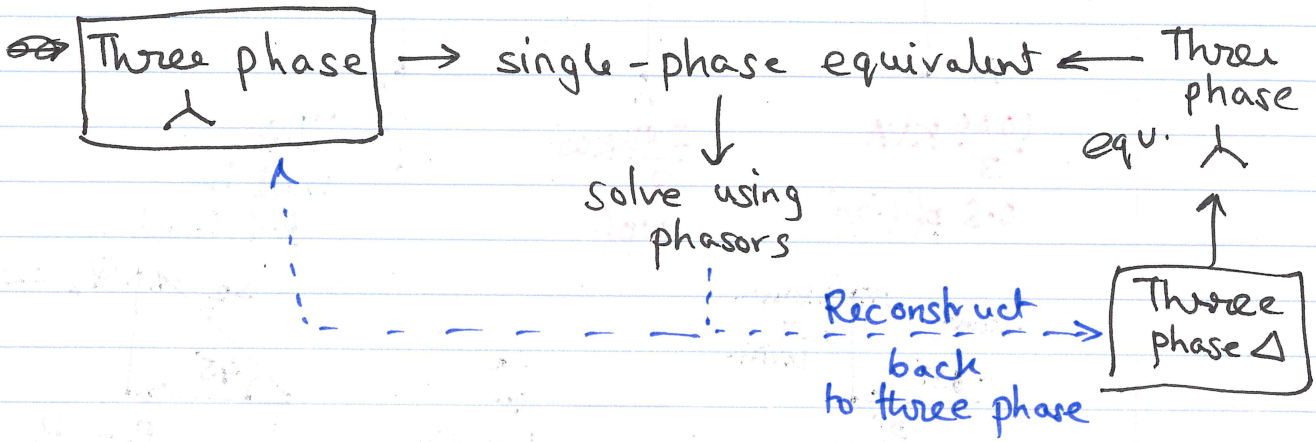
$$\bar{I}_A = \frac{\bar{V}_{AN}}{Z_2} = \frac{\bar{V}_{AB}}{\sqrt{3}\,Z_2}$$

$$\text{or,} \quad \frac{\bar{V}_{AB}}{\bar{I}_A} = \frac{Z_1}{\sqrt{3}}$$

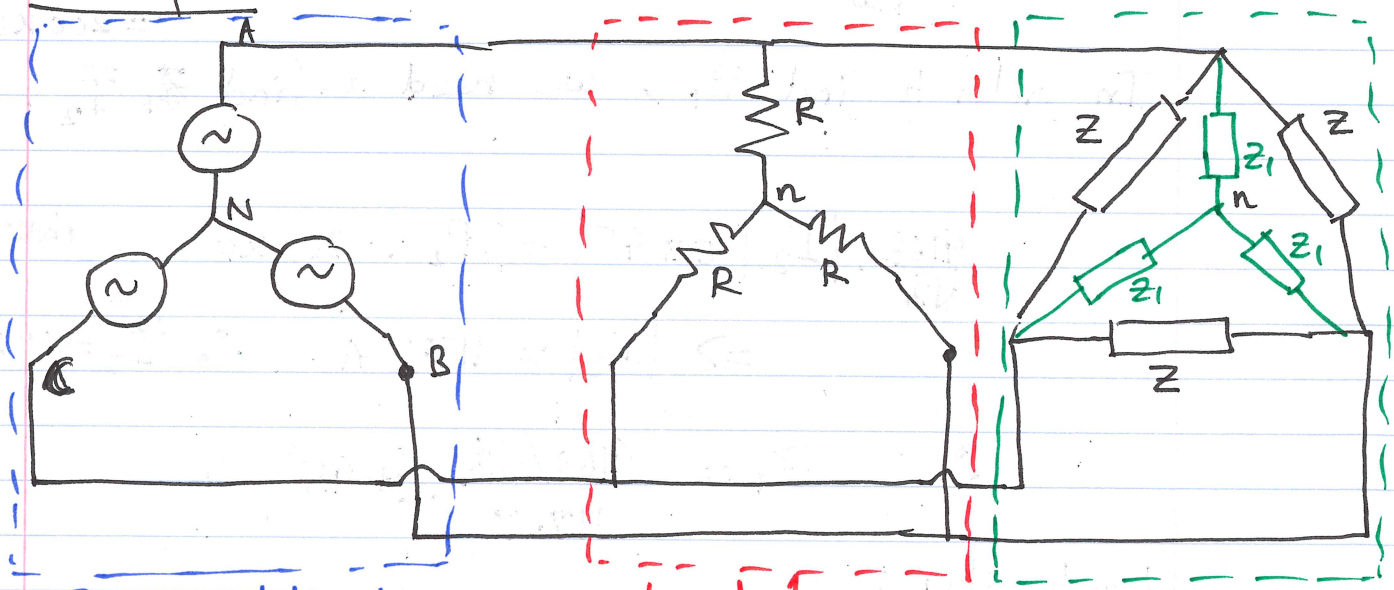
$$\frac{\bar{V}_{AB}}{\bar{I}_A} = \sqrt{3}\,Z_2$$

$$\therefore \boxed{Z_1 = 3Z_2}$$

B. So, how do we solve problems that involve delta configurations ?

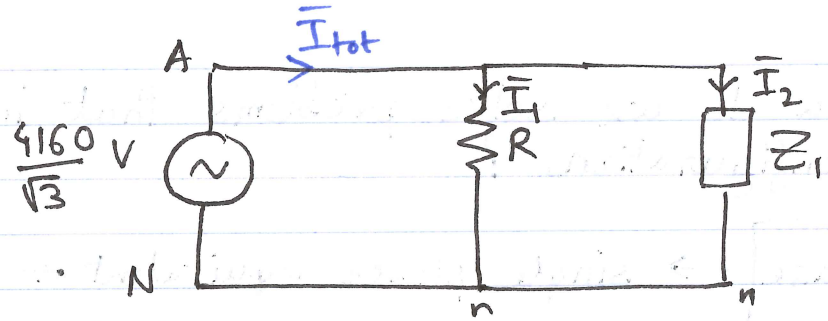


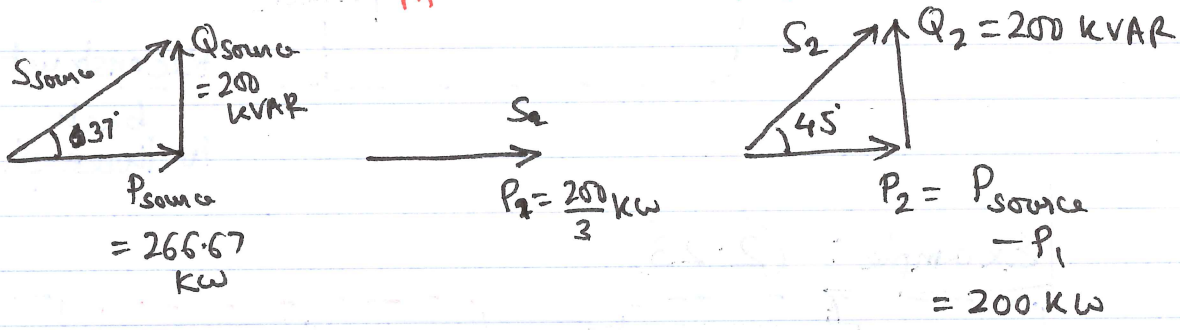
Example: (2.23)



Step 1: Reconstruct Δ-connected Z load 2 with equivalent Y-connected Z_1 load 2.

Step 2: Now you can draw single-phase equivalent





In order to find Z_1, we need to find $\bar{I}_2$.

$$\frac{4160}{\sqrt{3}} \cdot |\bar{I}_2| \cos 45^\circ = P_2 = 200 \text{ kW}$$

phasor

$$\Rightarrow |\bar{I}_2| = 117.7 \text{ A} \quad \therefore \bar{I}_2 = 117.7 \angle -45^\circ \text{ A}$$

lag.

$$\therefore \quad Z_1 = \frac{4160/\sqrt{3}}{117.7} \angle +45^\circ \ \Omega .$$

Last step:

$$\boxed{Z = 3Z_1 = 60.39 \angle 45^\circ \ \Omega}$$